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Integrated quality control project: Relationships between pathological findings detected at the slaughterhouse and information gathered in a veterinary health scheme at pig farms

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INTEGRATED QUALITY CONTROL PROJECT: RELATIONSHIPS BETWEEN PATHOLOGICAL FINDINGS DETECTED AT THE SLAUGHTERHOUSE AND INFORMATION GATHERED IN A VETERINARY HEALTH SCHEME AT PIG FARMS

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SUMMARY

In September 1986 an Integrated Quality Control Project (IQC) was started in the Netherlands. Over a period of 2 years a veterinary herd health scheme was applied to 21 growing and finishing pig herds. Information was collected concerning medical treatment, classified as group treatment and individual treatment, clinical observations, and housing factors. The main purpose of this project was to investigate relationships between pathological findings detected at the slaughterhouse and information gathered on the farm.

Medical group treatments, clinical observations and housing did not reliably predict pathological findings.

Individual medical treatment for respiratory disorders, started in the last 30 days before slaughter, proved to be positively related to the pathological findings 'pneumonia' and 'pleuritis' detected at the slaughterhouse. This may be useful for selecting pigs at the slaughterhouse that will require special treatment for the removal of the pleura parietalis.

During the first year of the project the average daily gain (ADG) was 27.2 g. more than during the preceding 2 years ($P < 0.0001$). Coughing was negatively related with the ADG. More than 75 coughs per compartment during 10 minutes, recorded more than 50 days before slaughter, was positively related to a depression of the ADG (20 g; $P < 0.05$).

Relationships between group medical treatments and ADG were statistically significant, when the group medical treatment was started more than 20 days before slaughter. Negative influences on the ADG were higher if the treatment was started closer to the slaughter date. Disregarding the reason for medical treatment, the depression of the ADG amounted to 23.8, 11.3, 3.9 and -5.9 g per day for intervals between medical treatment and slaughter of 21-40, 41-60, 61-80 and 81-100 days, respectively.

Blood samples were taken at slaughter from 20% of the slaughtered pigs. A total of 2039 blood samples were serologically tested for antibodies against *Actinobacillus pleuropneumoniae* (type 2 and 9) and influenza virus (type H1N1 and H3N2). Blood samples from pigs from all the farms showed antibodies against influenza H1N1, pigs from 95% of the farms had positive blood samples for *Actinobacillus pleuropneumoniae* type 2 and 9, while pigs from 86% of the farms had positive blood samples for Influenza H3N2.

INTRODUCTION

In September 1986 an Integrated Quality Control Project (IQC) was started in the Netherlands (3,4).

Data were gathered at the farm and at the slaughterhouse. The aim of the present study was to investigate relationships between pathological findings detected at the slaughterhouse and information gathered on the farm.

These data were collected in an Epidemiological Database System (EDB): the Veterinary Automated Management and Production control Program (VAMPP) for growing and finishing herds (2). Use of EDB's has been the subject of investigations during the last decade (4,5,6,7,13,22,23,24).

Relationships between the following groups of data were statistically evaluated in the present study:

- (i) group medical treatment and slaughterhouse data.
- (ii) clinical observations or housing factors and slaughterhouse data.
- (iii) individual medical treatment and slaughterhouse data.
- (iv) clinical observations or housing factors and Average Daily Gain (ADG).
- (v) group medical treatment and ADG.

Furthermore a serological survey of sera from 20% of animals at slaughter was performed. These sera were screened for *Actinobacillus pleuropneumoniae* type 2 and 9 and influenza virus (H1N1 and H3N2). On the basis of the frequency of detection of antibodies against one of these agents, the minimum sample size for detection of these infections was calculated.

MATERIALS AND METHODS

Farms

A veterinary herd health scheme was started in September 1986 (2). Twenty-one pig farms were selected on the basis of the following characteristics: at least 500 growing and finishing pigs kept in a compartment all in all out system and no reproduction area. Furthermore, the farmers had to deliver all the animals to one slaughterhouse and had to register technical data for at least 2 years before the IQC project. These herds participated in a integrated breeding, multiplying, fattening, and finishing program. Data were collected for 705 growing and finishing periods in different compartments of these farms over a period of 2 years.

All the pigs were vaccinated once against Aujeszky's Disease (by intramuscular or intranasal route) during the growing and finishing period. Pigs were not routinely vaccinated against APP spp. or Influenza spp.

Experimental design

Pigs in one pen were identified as a group by their location in a compartment. After individual medical treatment, a pig re-

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ceived an ear tag with a unique number. On the day before the day of transport to the slaughterhouse, each animal received another ear tag with another unique number. The generic names of drugs and the following information were noted:

- individual medical treatment :
medical treatment of one pig.
- group medical treatment :
medical treatment of all pigs in one pen with the same drug at the same day. When pigs in a compartment were treated, then this treatment was noted as a multitude of pen treatments.
- reason for medical treatment :
the organ system presumed to be affected was noted as were the clinical symptoms.
- drug frequency :
number of times that an active compound or drug was used.
- time :
the number of days between the start of the growing and finishing period and treatment.

The following clinical observations and housing factors were recorded at 4-week intervals :

- mean weight :
the estimated mean weight of the pigs.
- coughing and sneezing :
the number of coughs or sneezes observed in 10 min. per compartment, starting 5 min. after the clinical inspection of the pigs in the compartment.
- laboured breathing :
the number of animals per compartment showing clinical signs of laboured breathing.
- faeces :
three classes of consistency were noted: structured, gelatinous, and fluid.
- feed consumption per pen.
- tail biting wounds, ear biting wounds, lameness, underweight :
number of animals per pen and per compartment.

The following pathological findings were recorded at the slaughterhouse: Atrophic Rhinitis, pneumonia, pleuritis, abscesses in the lungs, skin lesions, inflammation of the tail, inflammation of a leg, arthritis, liver partially affected, liver condemned, and overfilled stomach. The definitions of these pathological findings are described in literature (9).

Slaughterhouse data, farm data, the results from the clinical inspection, and housing factors were the input data of the Veterinary Automated Management and Production Control Program for the growing and finishing herd (VAMPP) (2).

Statistical analysis

Relationships between pathological findings detected at the slaughterhouse and information gathered on the farm were investigated by using SAS (15,16,17,18). The percentage of animals showing a (certain) pathological finding was calculated per compartment and per pen. Data were evaluated with a General Linear Model (17,19). The model used was :
 $y = \text{constant} + a + b + c$

y: % pathological findings

a: medical treatment or clinical observation or housing factor;

b: correction for different growing and finishing periods in

time in the same compartment;
 c: correction for farm influences.

There was no transformation performed on the percentage of pathological findings, because the residual variation of the models did not improve when an arc-sinus transformation was used. As reference group, pigs were used from pens from the same farms, in the same period of time, and had not been given medical treatments in groups.

Based on the frequency of serologically positive pigs in the population, the minimum sample size for detection of *Actinobacillus pleuropneumoniae* and influenza virus H1N1 and H3N2 was calculated according to methods described in literature (1).

Serology

From October 1986 till March 1987 blood samples were taken at slaughter from 20 % of the animals. An haemagglutination inhibition test (HI) was used to screen animals on the farms for the prevalence of antibodies against influenza virus type H1N1 and H3N2 (10). A combined complement fixation test (CFT) was used to screen animals on the farms for the prevalence of antibodies against *Actinobacillus pleuropneumoniae* (App) type 2 and 9 (10). Reactions were considered positive for App or influenza virus when antibody titres were equal to or greater than 1:40 and 1:18, respectively (10).

If blood samples collected from two deliveries to the slaughterhouse had positive titres against either APP type 2 and 9 or influenza virus H1N1 or influenza virus H3N2, then no further serological screening was performed for antibodies against this agent.

RESULTS

Summary statistics

ADG per month ranged from 675 to 735 g, whereas the average mortality rate ranged from 0.6% to 2.0%. During the first year of the project, the ADG was 27.2 g higher than during the preceding 2 years ($P < 0.0001$). Strikingly, the frequency distribution of reasons for starting medical treatment was different for the group medical treatment compared with the individual medical treatment (Table 1).

Frequently observed were 'gelatinous faeces', 'ear biting',

Table 1. Frequency distribution of reasons for medical treatments

reason	group medical treatment			individual medical treatment		
	freq.	int.1	int.2	freq.	int.1	int.2
respiratory disorder	1869	33.9	66.1	1885	55.8	56.7
digestive tract disorder	3560	40.7	61.7	545	41.5	68.8
skin	1008	26.4	73.1	457	47.8	61.6
locomotion problems	17	17.2	77.8	1206	46.2	61.5

freq.: number of treatments

int.1: interval between starting date growing and finishing period and treatment date

int.2: interval between slaughter date and treatment date

Table 2. Relationships between clinical observations or housing factors and the frequency of pathological findings detected at the slaughterhouse. (p-observation / r^2 model).

code number	int.1	int.2	int.3	int.41
1	n.s.	n.s.	*/0.08	n.s.
2	n.s.	n.s.	*/0.03	n.s.
3	**/0.123	***0.09	n.s.	n.s.
4	n.s.	n.s.	n.s.	n.s.
5	n.s.	*/0.067	*/0.064	**/0.073
6	n.s.	*/0.067	*/0.065	*/0.072
7	n.s.	*/0.067	*/0.064	**/0.073
8	n.s.	***0.113	***0.141	***0.125
9	n.s.	***0.114	***0.142	***0.126
10	n.s.	***0.113	***0.142	***0.125
11	**/0.317	***0.238	***0.217	***0.177
12	***0.252	***0.187	***0.208	***0.175
13	***0.254	***0.183	***0.206	***0.173
14	***0.251	***0.186	***0.208	***0.172
15	***0.258	***0.199	***0.225	***0.177
16	***0.258	***0.216	***0.184	*/0.118
17	***0.252	***0.184	***0.213	***0.175
18	***0.31	***0.28	***0.27	***0.256
19	***0.316	***0.278	***0.239	***0.264
20	***0.313	***0.263	***0.235	***0.249
21	***0.3	***0.263	***0.235	***0.251
22	***0.313	***0.273	***0.237	***0.261
23	***0.329	***0.306	***0.284	***0.316
24	***0.304	***0.263	***0.236	***0.255
25	*/0.107	n.s.	**0.097	***0.077
26	*/0.103	n.s.	**0.088	***0.079

n.s.: $p > 0.05$ *: $p < 0.05$ **: $p < 0.01$ ***: $p < 0.0001$

int: interval between slaughter and observation date; int.1 ≤ 25 days,

int.2 > 25 ; ≤ 50 days, int.3 > 50 ; ≤ 75 days, int.4 > 75 ; ≤ 100 days

***: see list 1

List 1. Factors used in the relationships between clinical observations or housing factors and the frequency of pathological findings at the slaughterhouse (Table 2).

code number	slaughterhouse finding	clinical observation or housing factor
1	inflammation of the leg	lameness
2	arthritis	lameness
3	Atrophic Rhinitis	sneezing
4	inflammation of the skin	compartment temperature
5	inflammation of the tail	tail biting
6	liver partially affected	diarrhoea
7	liver partially affected	thin faeces
8	liver partially affected	underweight
9	liver condemned	diarrhoea
10	liver condemned	thin faeces
11	liver condemned	underweight
12	pneumonia	coughing
13	pneumonia	sneezing
14	pneumonia	tail biting
15	pneumonia	ear biting
16	pneumonia	compartment temperature
17	pneumonia	incoming air temperature
18	pneumonia	underweight
19	lungs impossible to mark	coughing
20	lungs impossible to mark	sneezing
21	abscesses in the lungs	coughing
22	abscesses in the lungs	sneezing
23	pleuritis	coughing
24	pleuritis	sneezing
25	pleuritis	compartment temperature
26	pleuritis	incoming air temperature

'coughing', 'sneezing', 'tail biting', 'lameness' and 'fluid faeces', which were reported in 34.1, 28.3, 24.4, 16.5, 16.4, 10.7 and 6.6% of the clinical observations, respectively.

Statistical analysis

Relationships between clinical observations or housing factors and the number of pathological findings detected at the slaughterhouse were investigated (table 2). The relationships that were statistically significant had an r^2 value from 0.03 to 0.329 (Table 2, list 1).

In addition, the relationship between clinical observations or housing factors and ADG was investigated. The clinical observation or housing factor frequently did not have a statistically significant influence ($p > 0.05$). Only coughing showed significant ($p < 0.0001$) relationships, with values ranging from 53.3 to 59.4. Coughing frequencies higher than 75 coughs per 10 min per compartment, observed between 50 and 75 days before slaughter, were related to a decrease of the ADG of 22 g. Coughing frequencies higher than 50 coughs per 10 min per compartment, observed between 75 and 100 days before slaughter, were related to a decrease of the ADG of 18 g.

Relationships between medical group treatments and pathological findings at the slaughterhouse were all statistically significant, with r^2 values ranging from 0.008 to 33.1. The r^2 increased with decreasing intervals between the day of medical treatment and slaughter.

Relationships between group treatments and ADG were statistically significant, when the group treatment was started more than 20 days before slaughter. That these relationships were not statistically significant when the group treatment was started less than 20 days before slaughter was probably caused by the low number of medical group treatments started during this period. With regard to intervals greater than 20 days, the negative influence on the ADG was higher with decreasing intervals between the day of treatment and slaughter. Disregarding the reason for medical treatment, the depression of the ADG amounted to 23.8, 11.3, 3.9 and -5.9 g per day for intervals between medical treatment and slaughter of 21-40, 41-60, 61-80 and 81-100 days, respectively.

Individual medical treatment for respiratory disease started in the last 30 days before slaughter was positively related to a significant increase in the pathological findings 'pneumonia' and 'pleuritis'. This increase in 'pneumonia' and 'pleuritis' increased with decreasing intervals treatment and slaughter.

Serology

Serological screening was done on 2039 blood samples obtained at slaughter from pigs from 21 farms. Antibodies against influenza H1N1 were found in samples from all the farms. On 20 farms pigs had antibodies against *Actinobacillus pleuropneumoniae* type 2 and 9. Pigs on 18 farms had antibodies against influenza H3N2. Antibodies against *Actinobacillus pleuropneumoniae* (type 2 and 9 combined) and influenza virus H1N1 and H3N2 were detected in 33.9%, 70% and 34% of the slaughtered pigs, respectively.

On the basis of the frequency of serologically positive pigs in the population, the minimum sample size for detection of *Actinobacillus pleuropneumoniae* and influenza virus H1N1

and H3N2 is 8, 2 and 8 pigs (95% confidence level), respectively.

DISCUSSION

The relationship between pathological findings detected at the slaughterhouse and clinical observations or housing factors proved to be statistically significant for almost all pathological findings (Table 2). However, the part of the variance that was explained by the relationships was not sufficiently high to enable a reliable prediction of the level of pathological findings at the slaughterhouse, based on clinical observations or housing factors. It seems that there are correlations in the residual variation of the models. The pattern of the residual variation was not influenced by an arc-sinus transformation.

Because there are statistically significant relationships, it might be possible to use this information in a veterinary herd-health management scheme. The relationships with the ADG could be used to improve the economics of the farms, while the relationships with the number of pathological findings detected at the slaughterhouse could be used to improve the quality of the products of the farm.

There were differences between the relative percentages of the reasons for starting medical treatment, when individual medical treatment and group medical treatment were compared. The reason for treatment 'digestive tract' was encountered in 53.5 % of the group medical treatments. On the other hand, the main reasons for individual medical treatment were 'respiratory tract' (39.7%) and 'locomotion' (25.4%). Relationships between medical group treatment and slaughterhouse findings were in all cases statistically significant. The part of variation in slaughterhouse findings that was explained by the relationships was not high enough to give a reliable prediction of the frequency of occurrence of slaughterhouse findings. Possible explanations for this phenomenon are:

- there is no consistency in the choice of giving a medical treatment.
- there is a difference in clinical symptoms of animals with the same slaughterhouse findings.
- the medical treatment is so effective that the percentage of slaughterhouse findings of treated pigs does not differ from that of untreated pigs.
- there is not enough consistency in the notation of slaughterhouse findings.
- not all the medical treatments are noted by the farmer.

Furthermore, the choice of the reference compartments or pens may have influenced the analysis. As reference group, the compartments or pens without medical group treatment were chosen instead of a control group of farms. This because the reference group should be under the same conditions as the medicated group.

Slaughterhouse findings and individual medical treatment were positively related, especially when the treatment was started during the last 30 days before slaughter, as were the slaughterhouse findings 'pneumonia' and 'pleuritis'. Individual medical treatment, started during the last 30 days before slaughter, irrespective of the reason for medical treatment, could be used to select pigs from which the pleura parietalis should be removed, because these animals had a higher incidence of 'pleuritis'.

Group medical treatment, started during the last 60 days before slaughter, was negatively related with the ADG. This is consistent with findings reported in the literature (12,14,20,21), and this seems to be indicative of underlying clinical problems.

Antibodies against influenza virus H1N1 and H3N2 were detected in 70% and 34% of the slaughtered pigs, respectively. These results differ from those of other investigations, whereby the percentage of animals with antibodies against influenza virus H1N1 and H3N2 ranged from 17.6 to 54.2% and from 30.3 to 70.8% respectively (8,11), but are in accordance with other reports (7). The minimum sample size for detection of animals with antibodies against *Actinobacillus pleuropneumoniae* type 2 and 9, influenza virus H1N1 and influenza virus H3N2 is based on the frequency of positive animals in the population. Thus the above discrepancy may have consequences for determining the minimum sample size for herd surveys.

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HAEMATOTOLOGY AND BIOCHEMISTRY REFERENCE VALUES FOR SOWS KEPT UNDER MODERN MANAGEMENT CONDITIONS

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SUMMARY

The data presented here were obtained using blood samples from 159 healthy, conventionally managed sows from 37 breeding herds. Sows were sampled at weaning and at 4-5 weeks gestation. Paired blood samples were analysed from sows that had a normal pregnancy and subsequently farrowed. The mean values of serum total protein, albumin and gamma globulin concentration were lower in the blood samples obtained at weaning compared with those obtained at 4-5 weeks gestation. Leucocyte count was higher, mostly as a result of a higher segmented neutrophil count, in the blood samples obtained at weaning compared with those obtained at 4-5 weeks gestation.

INTRODUCTION

Reference values for blood haematology and biochemistry are an important basis for the clinical interpretation of laboratory data (8). Blood reference values in swine generally relate to animals under experimental conditions and are based on a small number of animals (5, 10). Furthermore, much earlier work involved husbandry practices that are no longer current or methods were used that are now outdated, or swine breeds were used that are not representable for normal pig farming (7, 11, 14, 16). Swine husbandry has changed dramatically in the past two decades in the Netherlands, and laboratories have introduced automated systems with changes in methodology which have increased the precision and accuracy of blood chemistry measurements. Haematological and biochemical reference values for sows kept under modern management conditions in the Netherlands are therefore required to meet the changes in pig husbandry and laboratory technology. The whole concept of reference values and the normal range is controversial (6, 9). Despite the controversy, however, there is still a need for a range of normal or reference values. This article discusses some haematology and biochemistry reference values in sows at weaning and at 4-5 weeks gestation.

MATERIAL AND METHODS

Sows sampled

Thirty-seven swine breeding herds were selected from a database with 240 users of the computerized management software program CBK plus (13) in the Southern Netherlands (Table 1). To be selected, herds had to have complete recorded data from January 1 1987 to January 1 1988, and a herd size ≥ 80 sows. The herds were conventionally managed and sows were tethered. Herds used F₁ - sows (Finnish Landrace x Dutch Landrace). Sows were routinely vaccinated against

Table 1. Herd and productivity characteristics of the study population of 37 swine breeding herds in the Southern Netherlands with ≥ 80 sows per herd (January 1987 - January 1988).

	Study population	1661 reference herds in 1987 [#]
average herd size	180	130
average number of liveborn pigs/farrowing	10.4	10.5
average number of weaned pigs/farrowing	9.0	9.0
average number of weaned pigs/sow/year	19.4	19.6
average percentage of culled sows	44	48

[#] Baltussen et al. (1).

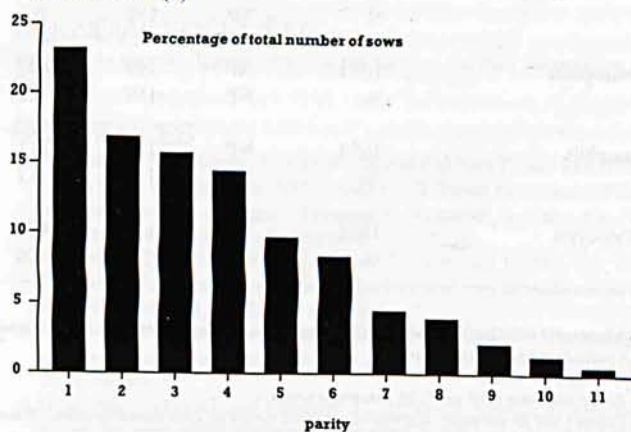


Figure 1. Distribution of parity for sows in the study (n=159).

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